

CHAPTER 7 - DEPLOYMENT

7.1 Introduction

Deployment is the phase in which the EduGenie application is prepared for execution and made available to users. The project was developed and tested locally using FastAPI and Visual Studio Code.

7.2 Development Environment

The application was built using Python, FastAPI, HTML, CSS, Google Gemini API, LaMini-Flan-T5, Git, GitHub, and Visual Studio Code.

7.3 Project Structure

The repository contains project source files, documentation, ER diagram, project workflow, screenshots, and the demo video. Organizing files into separate folders improves maintainability.

7.4 Running the Application

The application is executed using the command 'uvicorn main:app --reload'. After the server starts, users access the application through <http://127.0.0.1:8000> in a web browser.

7.5 GitHub Repository

GitHub is used for version control and project hosting. Source code, documentation, diagrams, reports, and other project assets are maintained in the repository.

7.6 Deployment Verification

Each module, including Question Answering, Explanation, Quiz Generation, Summarization, and Learning Path Recommendation, was tested after deployment to verify successful execution.

7.7 Advantages

Fast deployment, modular architecture, simplified maintenance, version control using GitHub, and scalability for future enhancements.

7.8 Chapter Summary

This chapter explained the deployment process, project organization, execution steps, GitHub usage, and verification of the EduGenie Learning Assistant.